

ensia	The National Higher School of Artificial Intelligence	2022/2023
	Computer Architecture II	Second Year

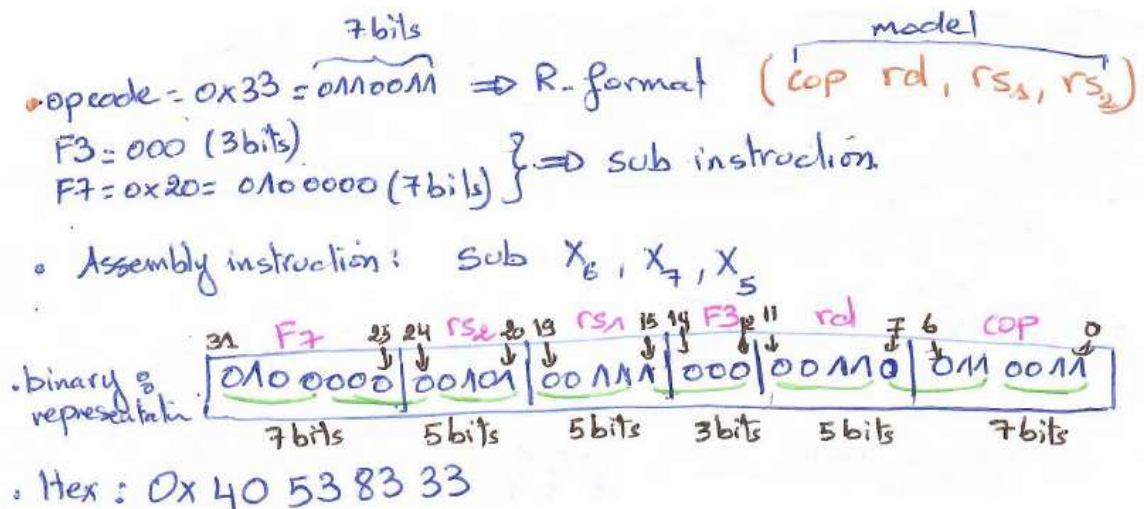
Assignment

Exercise 01

Provide the instruction type, assembly language instruction, and binary representation of instruction described by the following RISC-V fields:

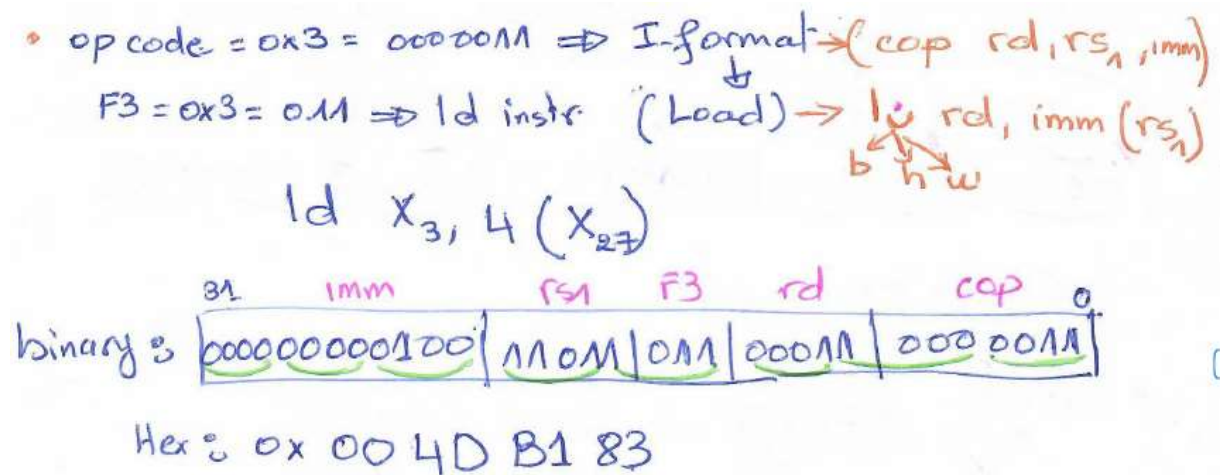
a) opcode=0x33, funct3=0x0, funct7=0x20, rs2=5, rs1=7, rd=6

R-type: sub x6, x7, x5 (0x40538333: 0100 0000 0101 0011 1000 0011 0011 0011)



b) opcode=0x3, funct3=0x3, rs1=27, rd=3, imm=0x4

I-type: ld x3, 4(x27) (0x4DB183: 0000 0000 0100 1101 1011 0001 1000 0011)



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Exercise 02

Considering the standard 32-bit RISC-V instruction formats, convert the following instructions to machine code

a) add x1, x2, x3

• add $\overset{rd}{x_1}, \overset{rs_1}{x_2}, \overset{rs_2}{x_3} \Rightarrow R\text{-format (0x33)}$

$\overset{F7}{|00000000|} \overset{rs_2}{|00011|} \overset{rs_1}{|00010|} \overset{F3}{|000|} \overset{rd}{|00001|} \overset{cop}{|0110011|}$

0x003100B3

a) addi x1, x2, 100

• addi $\overset{rd}{x_1}, \overset{rs_1}{x_2}, \overset{imm}{100} \Rightarrow I\text{-format (0x13)}$

$\overset{imm}{|000001100100|} \overset{rs_1}{|00010|} \overset{F3}{|000|} \overset{rd}{|00001|} \overset{cop}{|0010011|}$

$\Rightarrow 0x06410093$

$\overset{imm}{\uparrow} = 100_{10} = 000001100100_2$

(size): 12 bits

0x06410093

b) lb x1, 4(x2)

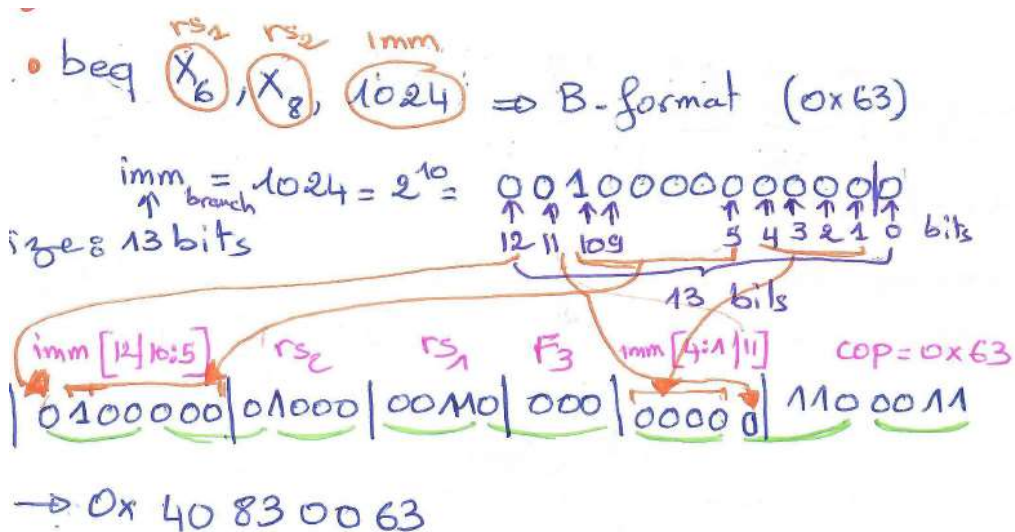
• lb $\overset{rd}{x_1}, \overset{imm}{4}(\overset{rs_1}{x_2}) \Rightarrow \text{load (I-format) (0x3)}$

$\overset{imm}{|0000000000100|} \overset{rs_1}{|00010|} \overset{F3}{|000|} \overset{rd}{|00001|} \overset{cop}{|0000011|}$

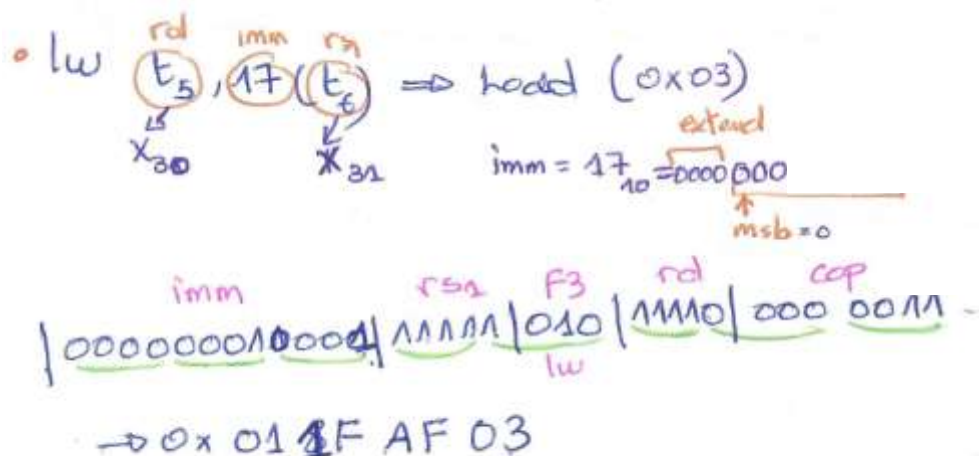
$\rightarrow 0x00410083$

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c) beq x6, x8, 1024



d) lw t5, 17(t6)

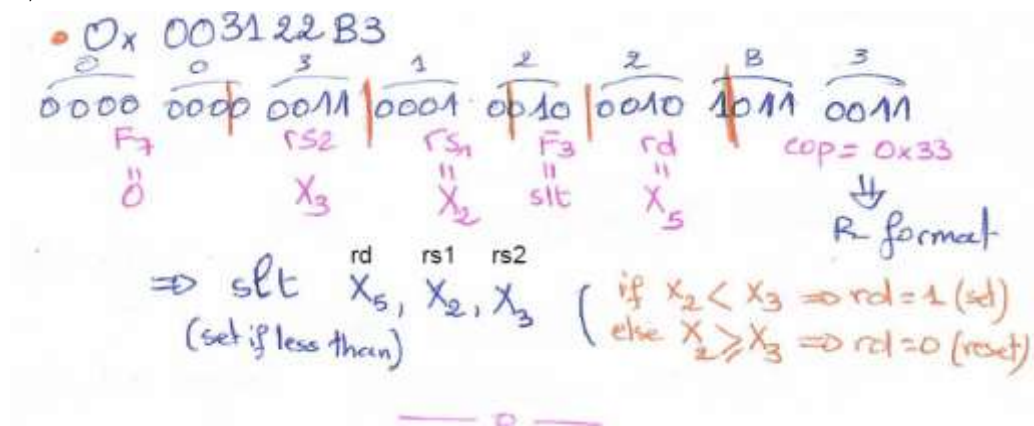


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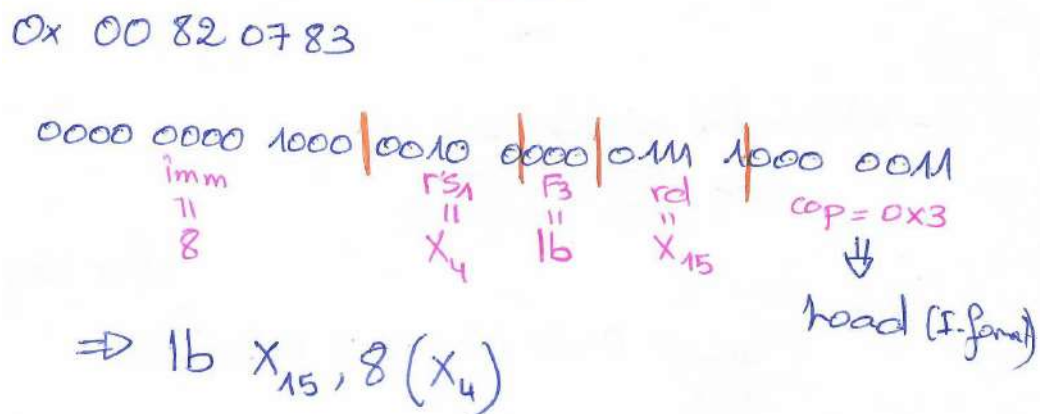
Exercise 03

What is the instruction encoded by

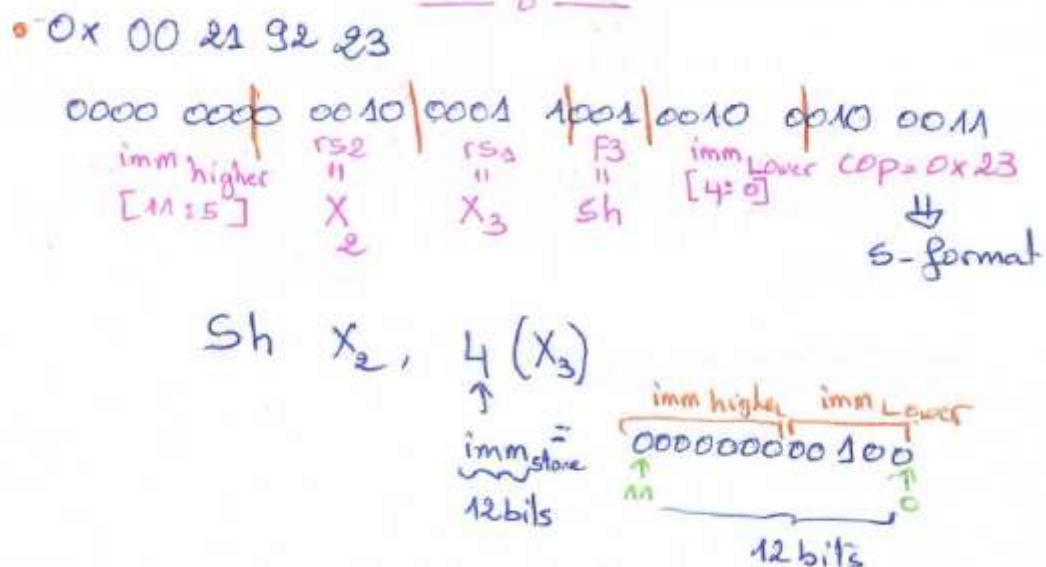
a) 0x003122B3



b) 0x00820783



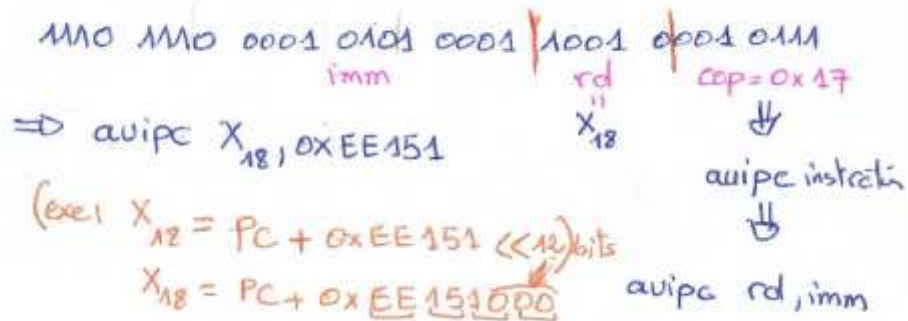
c) 0x00219223



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d) 0xEE151917

• 0x EE 15 19 17



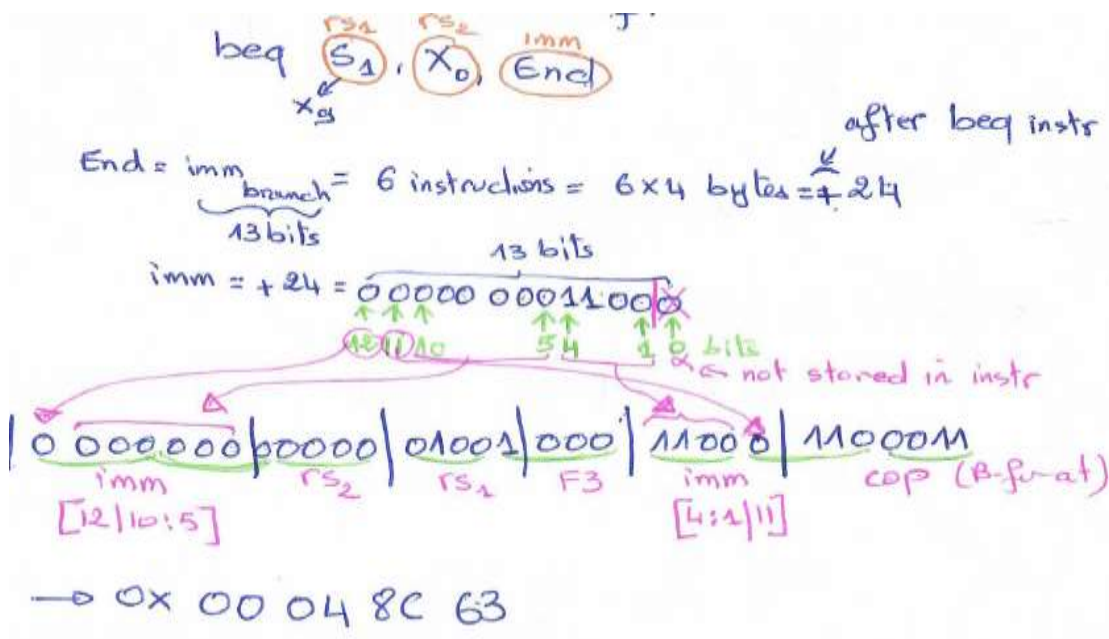
auipc x18, 0xEE151

Exercise 04

Consider the following assembly code in a file foo.s:

```
.text
1. mv s1 a0
2. addi s2 s2 4
3. Start: beq s1 x0 End
4. lw a0 0(s1)
5. jal ra printf
6. add s1 s2 s1
7. lw s1 0(s1)
8. jal x0 Start
9. End: jalr x0, ra, 0
```

a) What is the machine code generated for beq s1 x0 End (line 3).



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Exercise 05

You want to build a mini RISC-V instruction architecture that only supports 16 registers, which allows the length of the register fields to be shortened. Assuming that you use the extra bits to extend the immediate field, what is the range of half-word instructions that can be reached using a branch instruction in this new format? [<lower bound>, <upper bound>]

5 → Risc-V 32 (with 32 registers: $X_0 \rightarrow X_{31}$)

For the branch instruction, we have $\text{imm}_{\text{size}} = 13 \text{ bits}$ and each register is coded in 5 bits.

31 bits

imm [12 10 5]	rs2	rs1	rd	imm [4 1]	cop Br
7 bits	5 bits	5 bits	3 bits	5 bits	7 bits

Size of imm = 13 bits

Signed number ⇒ $\text{range} = [-2^{n-1}, 2^{n-1} - 1] \text{ bytes}$

⇒ $\text{range} = [-2^{12}, 2^{12} - 1] \text{ bytes}$ (byte addressable)

⇒ $\text{range} = [-2^{10}, 2^{10} - 1] \text{ instructions (4 bytes)}$

⇒ $\text{range} = [-2^{11}, 2^{11} - 1] \text{ Half Instructions (2 bytes)}$

In our exercise, we want to build a mini Risc-V with 16 registers ($X_0 \rightarrow X_{15}$)

0000 1111 4 bits

⇒ In this case, bits we only need 4 bits to encode registers.

imm	rs2	rs1	rd	imm	cop Br
4 bits	4 bits	3 bits	3 bits	4 bits	7 bits

18 bits

imm size = 14 bits + 1 bit (0)

15 bits

⇒ $\text{range} = [-2^{14}, 2^{14} - 1] \text{ bytes}$

⇒ $\text{range} = [-2^{12}, 2^{12} - 1] \text{ Instructions}$

⇒ $\text{range} = [-2^{13}, 2^{13} - 1] \text{ half Instructions}$

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Exercise 06

You are given the following RISC-V code:

```

Loop: andi t2 t1 1
      srli t3 t1 1
      bltu t1 a0 Loop
      jalr s0 s1 MAX_POS_IMM
      ...

```

1) What is the value of the byte offset that would be stored in the immediate field of the bltu instruction?

Two instructions away = -8 bytes

2) What is the binary encoding of the bltu instruction?

1111 1110 1010 0011 0110 1100 1110 0011

0xFE A36CE3